ROOF BOOK

ROOFTOP UNITS · FORMER AMARILLO STAR 14

ISSUED **27 AUG 2026**
PANEL READ **24 AUG 2026 06:30**
SHEETS **1.0 – 8.0**

PROJECT	SITE	SYSTEM	COMPILED FROM
POWER CHURCH Building activation	**8275 W AMARILLO BLVD** Amarillo, Texas	**NOVAR XCM.20R** NiagaraAX 3.5.34, orphaned	**4 SOURCES** 1998 design, 2023 + 2026 field, live panel

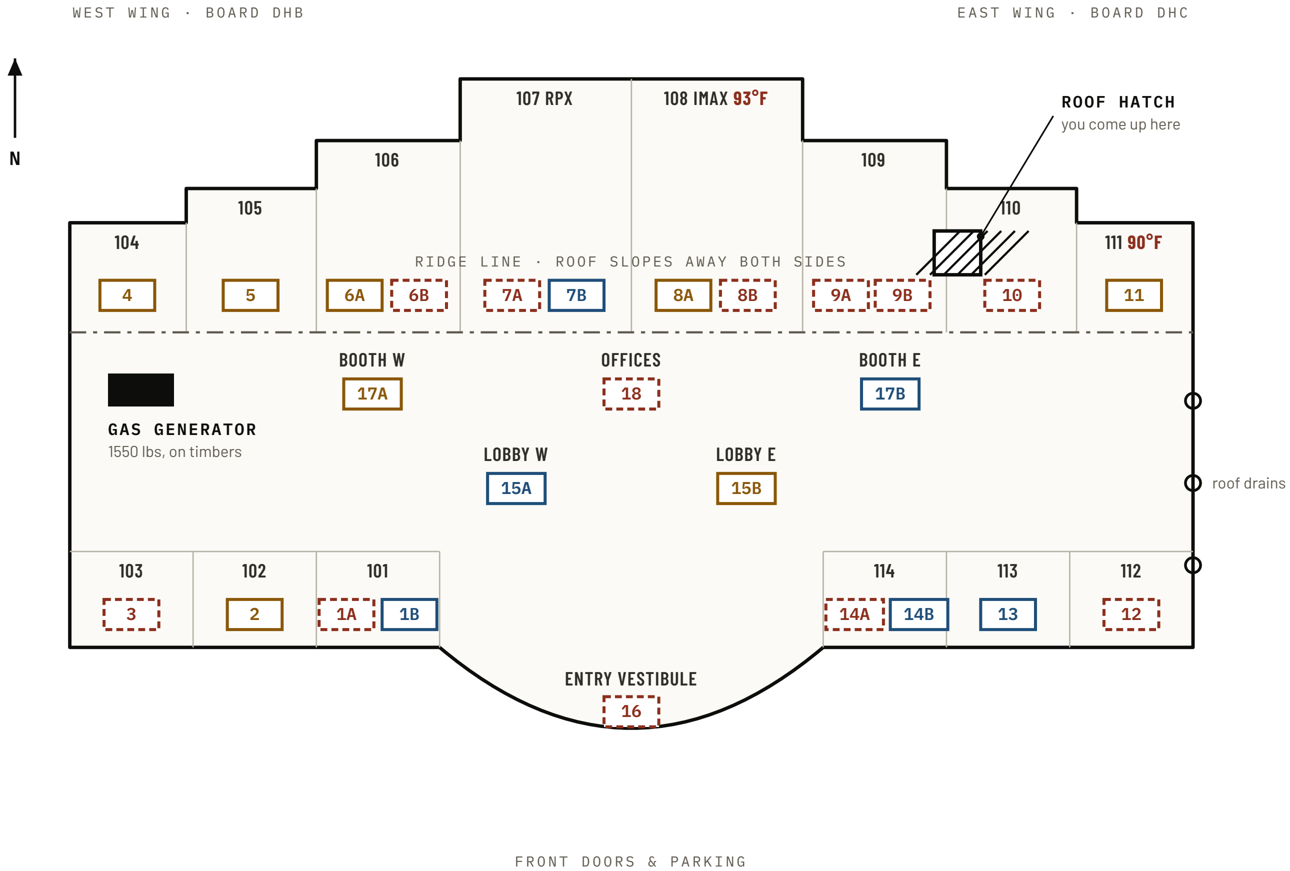
26	14	12	2	13
UNITS TOTAL ALL NOW IDENTIFIED	PANEL REACHES OBEY A COMMAND	NO COMMS HAND CONTROL ONLY	ROOMS OVER SETPOINT 108 IMAX · 111	PAST SERVICE LIFE 1998 LENNOX, 28 YRS

1.0 ROOF PLAN

Positions per 1998 sheet ME1.1. Schematic, not surveyed.

The building's own rule: **RTU-N sits directly over Auditorium 1NN.** RTU-3 runs Aud 103, RTU-11 runs Aud 111. Larger houses carry an A and a B unit. West wing feeds from board DHB, east wing from DHC.

● REACHES PANEL, FAN RUNNING ◑ REACHES PANEL, FAN IDLE ○ NO COMMUNICATION ▤ LANDMARK



Diagrammatic. Unit arrangement, hatch, ridge and generator follow 1998 sheet ME1.1; block widths are drawn for legibility, not to scale. **Standing at the hatch: the ridge runs left to right just in front of you, IMAX and RPX are the two big houses to your left, and the front doors are beyond them.**

2.0 UNIT SCHEDULE

Order parts from the field column, not the 1998 design.

Roughly half the roof was re-equipped with Trane units around the 2016 recliner conversion, at matching tonnage. The 1998 drawings still say Lennox for those slots and will send someone after the wrong part.

UNIT	SERVES	TONS	ON ROOF TODAY	BREAKER	PANEL	8/24	CONDITION AND REQUIREMENT
WEST WING — DISTRIBUTION BOARD DHB							
1A	Aud 101	4	Lennox 1998	2LA-37,39,41	NO COMMS	—	Bad blower 2023, still not running 2026. Replace.
1B	Aud 101	10	Trane 2016	DHB-2,4,6	RUNNING	75°	Healthy, cooling normally.
2	Aud 102	10	Trane / Am. Standard, newest	DHB-7,9,11	IDLE	70°	At target. Supply sensor reads 152°F with fan off — check sensor.
3	Aud 103	6	Lennox 1998	DHB-8,10,12	NO COMMS	—	Blower dead, compressor kicks. Replace.
4	Aud 104	6	Lennox 1998	DHB-13,15,17	IDLE	72°	Needs heat and cool work.
5	Aud 105	8.5	Lennox 1998	DHB-14,16,18	IDLE	71°	Bad combustion fan motor. Cools, will not heat.

UNIT	SERVES	TONS	ON ROOF TODAY	BREAKER	PANEL	8/24	CONDITION AND REQUIREMENT
6A	Aud 106	4	**Lennox 1998**	DHB-19,21,23	**IDLE**	71°	Fired clean 2023. No 2026 note.
6B	Aud 106	10	Trane 2016	DHB-20,22,24	**NO COMMS**	—	Healthy in 2023. Likely a comms fault, not a dead unit.
7A	Aud 107 RPX	7.5	**Lennox 1998**	DHB-43,45,47	**NO COMMS**	—	Contactor stuck, disconnect will not turn on.
7B	Aud 107 RPX	15	Trane 2016	DHB-25,27,29	**RUNNING**	77°	Cooling, supply 63°F.
15A	Lobby & promenade	20	Trane 2016	DHB-55,57,59	**RUNNING**	80°	Cooling. Large volume, slow pull-down.
17A	Projection booth	12.5	**Lennox 1998**	DHB-31,33,35	**IDLE**	74°	OA dampers damaged, tech left it switched off in 2023. Replace.
18	Offices & upper lobby	6	**Lennox 1998**	DHB-49,51,53	**NO COMMS**	—	Runs, but stuck and disconnected — will not turn off.
EAST WING — DISTRIBUTION BOARD DHC							
8A	Aud 108 IMAX	7.5	**Presumed Lennox 1998**	DHC-43,45,47	**IDLE**	93°	**Room hot, unit not cooling.** No field record from anyone, ever. Walk first.
8B	Aud 108 IMAX	15	Trane 2016	DHC-1,3,5	**NO COMMS**	—	Healthy 2023 and 2026, but invisible to the panel. IMAX has no working AC.
9A	Aud 109	4	**Lennox 1998**	DHC-2,4,6	**NO COMMS**	—	Bad blower motor since 2023. Replace.
9B	Aud 109	10	**Lennox 1998**	DHC-7,9,11	**NO COMMS**	—	Compressors kick, blower and heat do not. Loose panels. Replace.
10	Aud 110	8.5	**Lennox 1998**	DHC-8,10,12	**NO COMMS**	—	Cools, no heat, bad inducer motor. Replace.
11	Aud 111	6	Trane 2016	DHC-13,15,17	**IDLE**	90°	**Room hot.** Power at the board, will not run heat or cool. Newer unit — diagnose before replacing.
12	Aud 112	6	Trane 2016	DHC-14,16,18	**NO COMMS**	—	Heat and cool work. Electrical door missing, everything got wet. Likely why comms died.
13	Aud 113	10	Trane 2016	DHC-19,21,23	**RUNNING**	76°	Best unit on the roof. Heat and cool both confirmed.
14A	Aud 114	4	**Lennox 1998**	DHC-20,22,24	**NO COMMS**	—	Fired clean 2023. Likely comms, not the unit.
14B	Aud 114	10	Trane 2016	DHC-25,27,29	**RUNNING**	76°	Supply air 51°F. Working exactly as intended.
15B	Lobby & promenade	20	Trane 2016	DHC-49,51,53	**IDLE**	80°	Sitting at its 80°F setback target, which is why it is not running.
16	Entry vestibule	12.5	**Lennox 1998**	DHC-55,57,59	**NO COMMS**	—	Needed transformer and fuses 2023. **By 2026 no power at all.** Same corner as the dead lobby lighting circuit.

UNIT	SERVES	TONS	ON ROOF TODAY	BREAKER	PANEL	8/24	CONDITION AND REQUIREMENT
17B	Projection booth	10	Trane 2016	DHC-31,33,35	**RUNNING**	79°	Blower runs, errors on heat and cool. Supply reading 31°F suggests a sensor fault.

3.0 UNIT IDENTIFICATION

The stencils faded. Three of four recovered from paper.

Texas Air found several units with no readable tag in July and described them by position: *"the Lennox between RTU-1B and the American Standard."* That reads like a dead end. The 2023 walkthrough had recorded **serial numbers** for the same units without knowing their positions. Matching the two lists supplies both halves.

3.1 **"Lennox between RTU-1B and the American Standard" = RTU-1A.** Serial 5699D02422 matches the untagged 4-ton Lennox logged in 2023 with a bad blower, and position confirms it — 1A is 1B's pair. The American Standard beside it is **RTU-2**, the newest unit on the roof.

3.2 **"Trane between RTU-12 and RTU-11" = RTU-17B.** Serial 185015437L is an exact match to the projection booth unit, which sits in that cluster on the roof plan.

3.3 **"Lennox in the middle in front of RTU-10" = RTU-9A.** Model GCS20-048-120-1G matches exactly, and 9A sits beside RTU-10 — the roof hatch is drawn between them.

3.4 **"Lennox in front of RTU-13", model REMD16M-65-2 — still unidentified.** Appears in no schedule we hold. Likely one of the split-system condensers serving the ticket booths and workroom rather than a rooftop unit. Needs eyes on the nameplate.

> **PERMANENT FIX**
>
> Sheet note 11 of the 1998 set required every unit stenciled in 2-inch black letters with its number and the area served. That stencil is gone. **Re-stencil all 26 on the first roof walk** and this problem never returns. It is a can of paint.

4.0 OPERATION

Three routes, depending on the unit.

4.1 **From a phone, for the 14 units the panel reaches.** Get on the panel's local network, open the dashboard, press Start Cooling. Runs one hour, reverts on its own. This is the panel's designed after-hours override, so units keep their normal compressor safeties.

4.2 **From the panel itself, same 14 units.** Browse to `http://192.168.1.123` and sign in. The panel's built-in screen is a dead Java applet and shows blank; the working route is the data interface behind it, which is what the dashboard uses.

4.3 **At the unit, for the 12 the panel cannot reach.** These will not answer any command from the office. Someone goes to the roof and uses the disconnect on the unit. This is the entire reason the no-comms list matters.

LIMIT

A cooling command only reaches the 14 units that are talking. Aud 108 is the clearest case: 93°F, one unit idle and the other invisible to the panel. No amount of pressing buttons cools that room today.

5.0 OCCUPANCY SCHEDULES

All five still on the theater's operating day.

The panel runs five named schedules. Every one is still set to Regal's movie hours, which is why rooms sit in setback when you expect them cool.

STORE SCHEDULE

OPENS 11:00

The master day. Last edited March 2022.

EMPLOYEE SCHEDULE

OPENS 10:30

Shadows the store day, offset earlier. Edited 16 Aug 2026, not by us — worth asking who.

AUDITORIUM EMPLOYEE

OPENS 12:30

Untouched since July 2015.

EXTERIOR SCHEDULE

OPENS 17:30

Parking and exterior lighting. Edited Aug 2022.

BATTERY PACK / EMERGENCY

ALWAYS OPEN

Effectively always on. Next change not until November.

6.0 CONTROL REPLACEMENT PLAN

Two live paths. They cover different units.

The panel's weekly calendar can only be edited with Niagara Workbench, which is version-locked to this generation and sold only to certified contractors. We asked the one person best placed to obtain it.

KEITH SVONOVEC, BGIS — 25 AUG 2026

"AX Novar is no longer. Novar won't budge. I tried previously."

He already tried on his own account and was refused. The official route to the schedule is closed. Nobody spends another dollar or another week on it.

6.1 THE RASPBERRY PI — FOR THE 14 UNITS THAT STILL ANSWER

The plan does not fight the software. **It moves the schedule off the panel entirely.** A small computer holds the church's real hours and presses the same button a person would press, on a clock instead of by hand. Regal's five schedules stay untouched underneath.

ANYONE, ANY PHONE

CHURCH WIFI

Open a bookmark. No app, no panel login, no laptop plugged into anything.

↓

THE BRIDGE

RASPBERRY PI

Two connections at once: WiFi to the church, cable to the panel. Holds the dashboard, the password, and the hours file.

↓

192.168.1.123

NOVAR PANEL

Receives the same override we tested by hand and passes it to the 14 units it can reach.

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a **A plain hours file.** One small file lists the days and times cooling should run. Editing it is editing a list, not programming a building.

b **The Pi checks the time once a minute** and works out whether the building should be cooling or coasting.

c **It only acts on a change.** When the answer flips it sends the identical command the Start Cooling button sends. It does not re-send every minute, so the panel is never spammed.

d **The Pi becomes the real schedule.** Church hours live there. Regal's movie hours stay dormant underneath, harmless, never edited.

THREE LIMITS BEFORE ANYONE COUNTS ON IT

Turning cooling on is proven. Turning it off is not yet. We fired the on command against the real panel 23 Aug and watched units engage. The off path uses a value not yet tested on live equipment. We learn at the first real transition, with someone watching.

Whole-building, not per-room. Same as the manual button. It cannot schedule Aud 109 differently from the lobby.

The lighting buttons will look like they work and won't. Keith's read is this building never had much interior lighting on the panel to begin with, so it is not a fault awaiting repair.

6.2 STATUS AND THE TWO FIXES

Code written and reviewed. No physical Pi built yet. Hardware is roughly **$165 once** for a complete Pi 5 kit; WiFi and ethernet are built in, so no adapters.

i **The network config needs replacing.** Written for the older dhcpcd system, but current Raspberry Pi OS uses NetworkManager, so the static address would silently never apply.

```
nmcli connection add type ethernet ifname eth0 con-name panel-eth0 ipv4.addresses 192.168.1.51/24
ipv4.method manual ipv4.never-default yes
```

ii **Give the Pi .51, not .50.** A laptop plugged in at the same time would collide on .50, which is what we used on 23 Aug.

iii **Add a PIN before the link travels.** Fine unlocked while it is only reachable on church WiFi. The moment it is forwarded to the internet it needs a login first.

Worth saying plainly: the Pi build is **better than what we ran on 23 Aug**, because the panel password lives only on the Pi and never sits in anyone's phone browser.

6.3 THERMOSTATS — FOR THE 12 THE PANEL CANNOT REACH

Keith's recommendation, and it is incremental rather than all at once: *"start to put RTU's on thermostats as they go offline."* Roughly **$150–350 per unit plus labor**, commercial multi-stage, not a home thermostat. Sam Patton has asked him to name a specific WiFi or BACnet model; that answer is pending.

6.4 CLOSED. DO NOT REOPEN THESE IN A MEETING.

× ~~OLD AX WORKBENCH~~

Does not exist. Keith tried to obtain it himself and Novar refused.

× ~~CURRENT NIAGARA WORKBENCH~~

Cannot be purchased without a certification number, and the current version targets migrating off our generation rather than running it.

× ~~REMANUFACTURED EMS CONTROLLERS, $350–500 EACH~~

Buys nothing. **Restoring a unit's communication does not give the schedule back** — the schedule is locked behind software we cannot get, not behind the comms. Same position, minus the money.

× ~~MIGRATING THE PANEL TO N4~~

Wrong direction. Thousands to modernize the brain of a system whose unit controllers are obsolete and which we are already leaving one thermostat at a time.

THE PLAN IN ONE SENTENCE

The Pi runs the schedule for the 14 units that still answer. Thermostats take over each of the other 12 as we get to them. The panel keeps doing the one job it is still good at, and we stop trying to win back the part that is gone.

7.0 FIELD NOTES

Keith Svonovec, BGIS — 25 Aug 2026

Keith has serviced this exact panel. His notes change three things we were treating as unknown and hand us one practical trick.

THE 12 OFFLINE UNITS MAY BE SELF-REGULATING

"If still on EMS and not communicating but controlled by EMS, I think the backup setpoints are 74 cooling and 67 heating."

A unit cut off from the panel can still hold roughly **74°F on its own internal fallback**. It is not necessarily doing nothing. He hedged with "I think," so treat it as a lead to verify — testable for free with a thermometer in rooms served only by offline units.

One room already argues against it. Aud 108 sat at 93°F with its only other unit offline. If that fallback were holding, the room would not be at 93°F. So either it is not running there, or that unit has genuinely failed.

7.1 **Check whether some units already have thermostats.** A possibility nobody had considered: some of the 12 may read as offline because their EMS controllers were already removed and thermostats fitted at some point. If so, those units are not broken and need nothing. Free to check on the roof walk, and it could shrink the problem list on the spot.

7.2 **Verify power at the unit first.** His first diagnostic before anything else. RTU-16 is the proof case — no power at all.

7.3 **The wiring trick that saves real money.** Each RTU already has a wire running down to its room. Use that existing wire as a **pull string** for new thermostat cable, since the old wire almost certainly lacks enough conductors. That turns a conduit-fishing job into a pull-through.

7.4 **Or mount the stat at the existing sensor and lock it.** Every auditorium already has a wall sensor at 72 inches by the vestibule door. Putting the thermostat there avoids new wire entirely; lock it with software or a plastic guard.

7.5 **The interior lighting mystery is explained, not broken.** *"Lighting most likely just exterior was done in the past. Many of the old theatres didn't have much interior lighting control."* That matches what we found live. Stop treating it as a fault to repair.

8.0 WORK LIST

All five happen on one roof walk.

Take a thermometer, a camera, a can of stencil paint, and this page.

8.1 **Walk RTU-8A first.** It runs IMAX, the hottest room in the building at 93°F, and it is the one unit with no inspection record from anybody, ever.

8.2 **On each of the 12 offline units, answer Keith's two questions.** Does it have power? Does it still have its EMS controller, or did someone already fit a thermostat? Either answer can remove a unit from the list for free. Several were mechanically healthy at last inspection — 6B, 8B, 12 and 14A all fired clean.

8.3 **Take room temperatures while you are there.** Rooms served only by offline units tell you whether the 74°F fallback is actually holding. That single reading decides how urgent the thermostat spend really is.

8.4 **Re-stencil all 26.** A can of paint permanently ends the identification problem that cost two field visits and a week of paperwork.

8.5 **Trace the entry-vestibule power.** RTU-16 has no power and the lobby's left-wall lighting circuit is dead. Two independent findings, same corner, months apart. One electrician visit likely explains both.

CAPITAL EXPOSURE

The real question is not "replace the HVAC." It is **replace the 13 remaining 1998 Lennox units, on a schedule.** The 12 Trane units are about 11 years old with a decade left, and not one of them had a fault in 2023. At Texas Air's own stated ballpark of $25k–$35k per unit, those 13 represent roughly **$325k–$455k**. That is a planning figure from a service invoice, not a bid. Get real quotes before it goes near a lender.

WHERE A CHURCH CONNECTION ACTUALLY SAVES MONEY

Diagnosing comms faults, replacing a transformer and fuses, pulling thermostat wire and re-stenciling are ordinary HVAC trade work — a congregant's crew can do all of it. Programming the Novar panel is a different, specialized trade. Split the ask that way and volunteer help lands where it genuinely helps.

SOURCES 1998 EnviroDesign / Stotser MEP set, sheets M4.1 and ME1.1 — 26-unit schedule, roof positions, breaker assignments · Allen's Tri-State Mechanical roof walkthrough Nov 2023 — make, age, serial, fired or not · Texas Air invoice 11419, 9 Jul 2026 — current condition, replacement ballpark · Novar XCM.20R panel read live 24 Aug 2026 06:30 — comms status, zone temperature, setpoint, fan state · Keith Svonovec, BGIS, correspondence 21 and 25 Aug 2026

NOTE Live temperatures are one snapshot taken during an active cooling override, not a steady-state reading. Unit identities in sheet 3.0 are inferences from matching serial numbers and models across two independent field reports; they are strong but should be confirmed against the nameplate on the first roof walk.